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AIM

To study the current–voltage (V–I) characteristics of a Gunn diode, to locate the threshold (peak) voltage, and to identify the negative–resistance region of the device.

EQUIPMENT AND COMPONENTS

- 1 Gunn power supply.
- 2 Gunn oscillator with Gunn diode mount.
- 3 PIN modulator.
- 4 Isolator.
- 5 Frequency meter.
- 6 Variable attenuator.
- 7 Detector mount with matched crystal detector.
- 8 Waveguide stands.
- 9 VSWR meter / Cathode Ray Oscilloscope.
- 10 BNC connecting cables and cooling fan.

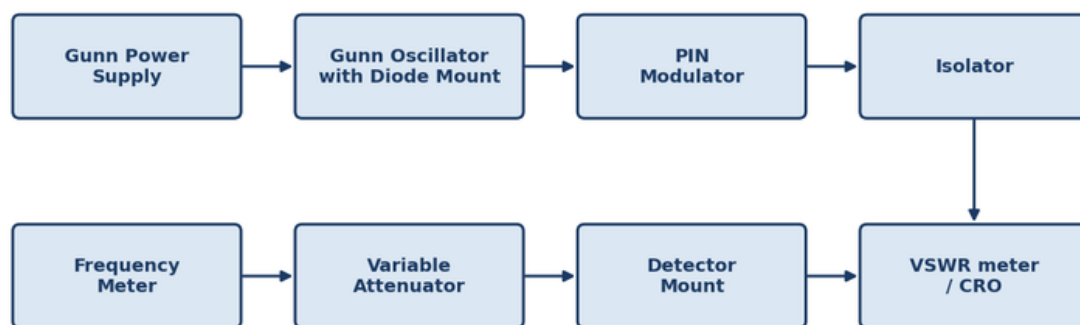
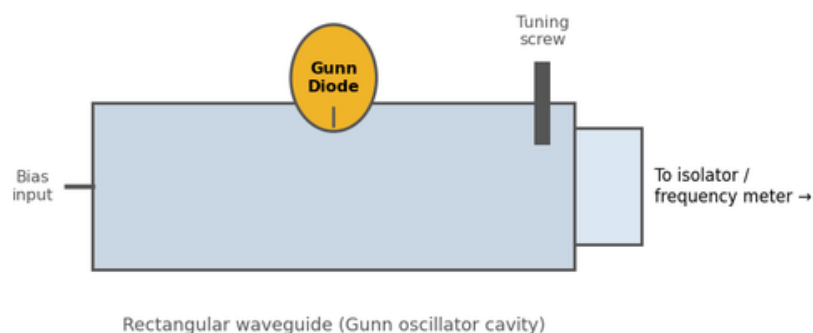
EXPERIMENTAL SETUP

Fig. 1 — Microwave bench arrangement for Gunn diode V–I characteristics

Gunn Oscillator with Diode Mount (schematic)

Photographic/schematic view of the Gunn oscillator with diode mount used in the bench

THEORY

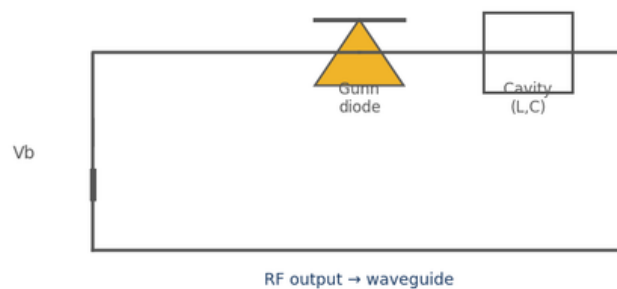
The Gunn diode is a two–terminal, negative–resistance active device made from a bulk semiconductor such as Gallium Arsenide (GaAs) or Indium Phosphide (InP), which does not use a p–n junction. Its operation is based on the Gunn effect, also called the transferred–electron effect. In these compound semiconductors, the conduction band has two energy valleys: a lower valley in which electrons have high mobility, and an upper satellite valley in which electrons have low mobility.

At low applied electric fields, nearly all the conduction electrons stay in the high-mobility lower valley and the current increases linearly with voltage, exhibiting normal ohmic behaviour. As the field is raised beyond a critical (threshold) value, a significant fraction of electrons gain enough energy to be scattered into the upper, low-mobility valley. Because their effective mobility now falls even though the field keeps increasing, the drift velocity – and hence the current – decreases with increasing voltage. This produces a region of negative differential resistance in the device characteristic. Beyond this region, high-field domains form and travel through the diode, giving rise to a periodic current oscillation whose frequency depends on the transit time of the domain across the active layer, which is the basis of Gunn oscillation in the microwave range.

The complete V-I curve of a Gunn diode therefore consists of three broad regions: an initial ohmic rise, a negative-resistance region beyond the threshold (peak) voltage where current falls as voltage rises, and a saturation region at higher voltages.

CIRCUIT / PORT DIAGRAM

Gunn Diode Bias Circuit (equivalent representation)

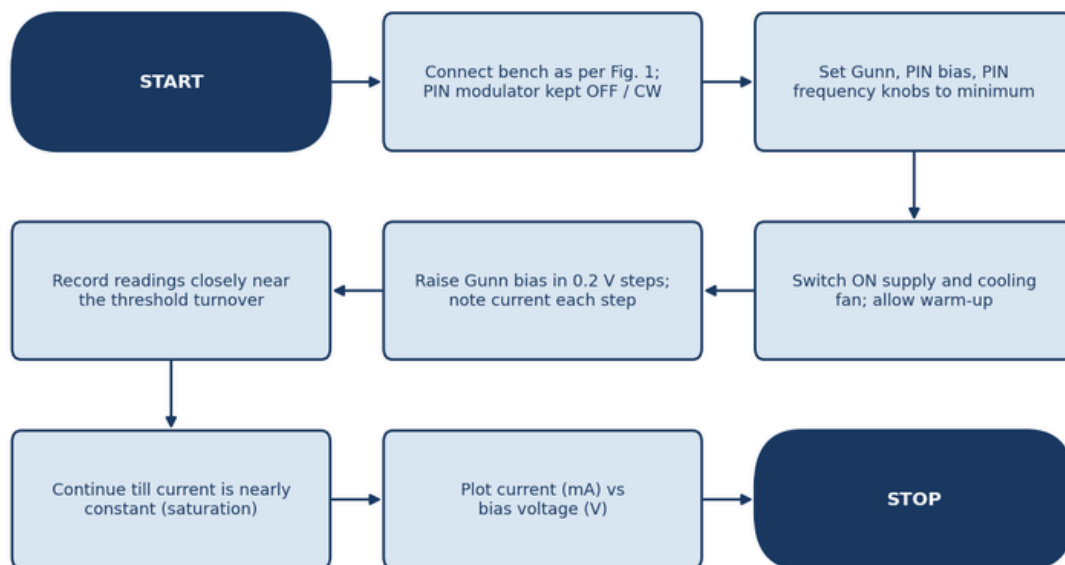


Equivalent bias circuit of the Gunn diode showing the negative-resistance element coupled to the resonant cavity

PROCEDURE

1. The equipment is connected as shown in Fig. 1, with the PIN modulator kept in the “OFF”/CW position so that only the diode bias is varied (no modulation is applied for this test).
2. All the control knobs on the Gunn power supply (Gunn bias, PIN bias, PIN frequency) are initially set to the minimum position.
3. The Gunn power supply and the cooling fan are switched ON, and a short warm-up time is allowed.
4. The Gunn bias voltage knob is increased in small steps (of about 0.2 V), and the corresponding meter current is noted at every step.
5. Readings are taken carefully around the region where the current stops rising and begins to fall, since this identifies the threshold voltage.
6. The bias voltage is increased further, in the same steps, until the current again becomes nearly constant (saturation), and readings are recorded throughout.
7. A graph of diode current (mA) versus bias voltage (V) is plotted from the tabulated readings.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp-01

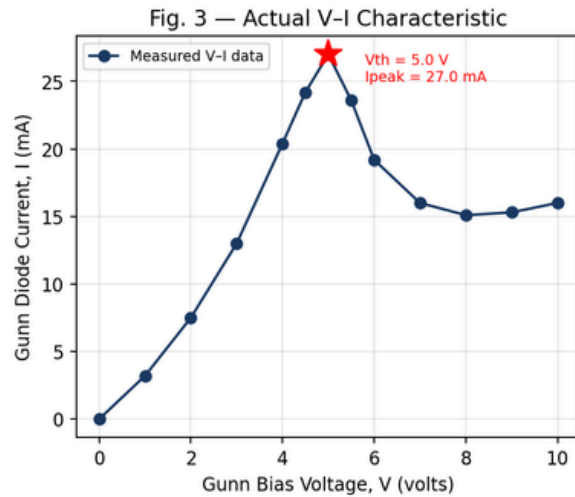
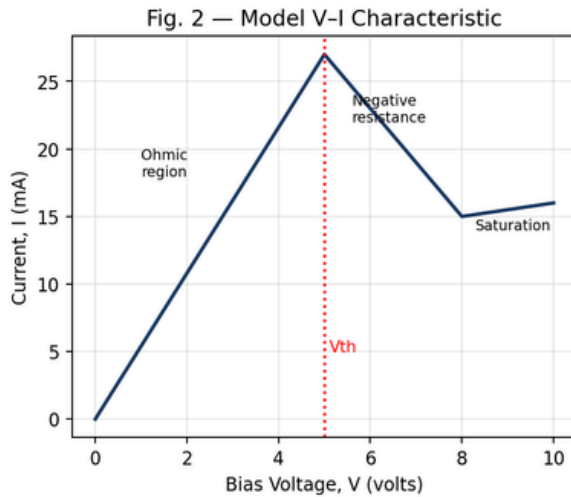
TABULATION

S. No.	Gunn Bias Voltage (V)	Gunn Diode Current (mA)
1	0.0	0.0
2	1.0	3.2
3	2.0	7.5
4	3.0	13.0
5	4.0	20.4
6	4.5	24.2
7	5.0 (V _{th})	27.0 (I _{th} , peak)
8	5.5	23.6
9	6.0	19.2
10	7.0	16.0
11	8.0	15.1
12	9.0	15.3
13	10.0	16.0

FORMULA

1. Threshold field, $E_{th} = V_{th} / d$, where d is the active-layer length of the diode (taken as $d \approx 15.2$ microns (μm) for this GaAs Gunn diode).

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. From the tabulated readings and the plotted graph, the current increases with bias voltage up to a maximum value of 27.0 mA at 5.0 V. Beyond this point the current falls with increasing voltage up to about 8.0 V, after which it levels off (saturation region).
2. Threshold voltage, $V_{th} = 5.0 \text{ V}$ (from graph, at $I_{peak} = 27.0 \text{ mA}$).
3. Negative-resistance region: $5.0 \text{ V} \leq V \leq 8.0 \text{ V}$ ($\Delta I / \Delta V < 0$ in this range).
4. $E_{th} = V_{th} / d = 5.0 \text{ V} / (15.2 \times 10^{-4} \text{ cm}) \approx 3.3 \times 10^3 \text{ V/cm} = 3.3 \text{ kV/cm}$, which agrees with the standard threshold field quoted for bulk GaAs.

PRECAUTIONS

1. All bias knobs must be kept at the minimum position before switching the power supply ON.
2. The Gunn bias voltage must never be increased beyond the safe operating limit specified for the diode (typically not more than about 10 V), to avoid damaging the device.
3. Readings should be taken only after allowing sufficient warm-up time.
4. Connections should be checked for tightness before switching ON.
5. Do not look directly into the open end of the waveguide.
6. The cooling fan must remain ON throughout the experiment.
7. After completing the experiment, all knobs should be returned to the minimum position before switching OFF.

RESULT / CONCLUSION

The V-I characteristics of the given Gunn diode were studied and plotted. The threshold (peak) voltage was found to be **$V_{th} = 5.0 \text{ V}$** , corresponding to a peak current of **27.0 mA** , and the device exhibited a clear negative-resistance region between **5.0 V and 8.0 V** , confirming the transferred-electron (Gunn) effect.